



CARE-PCM: Classification and Analysis of Reliability with Efficient Models for PCM

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Motivation

- Phase Change Memory (PCM) offers scalability, high density, and DRAM-like speed.
- However, PCM has limited write endurance ($\sim 10^8$ cycles), leading to degradation.
- Predicting PCM health is critical to ensure system stability and reliability.
- Goal: Develop lightweight, interpretable ML models for real-time PCM monitoring.

Research Objective

- Classify PCM cells into: Healthy, Degraded, and Near-Failure.
- Use operational indicators: write count, error frequency, and temperature.
- Compare Decision Tree and Logistic Regression models.
- Aim for efficient and interpretable real-time reliability monitoring.

Background

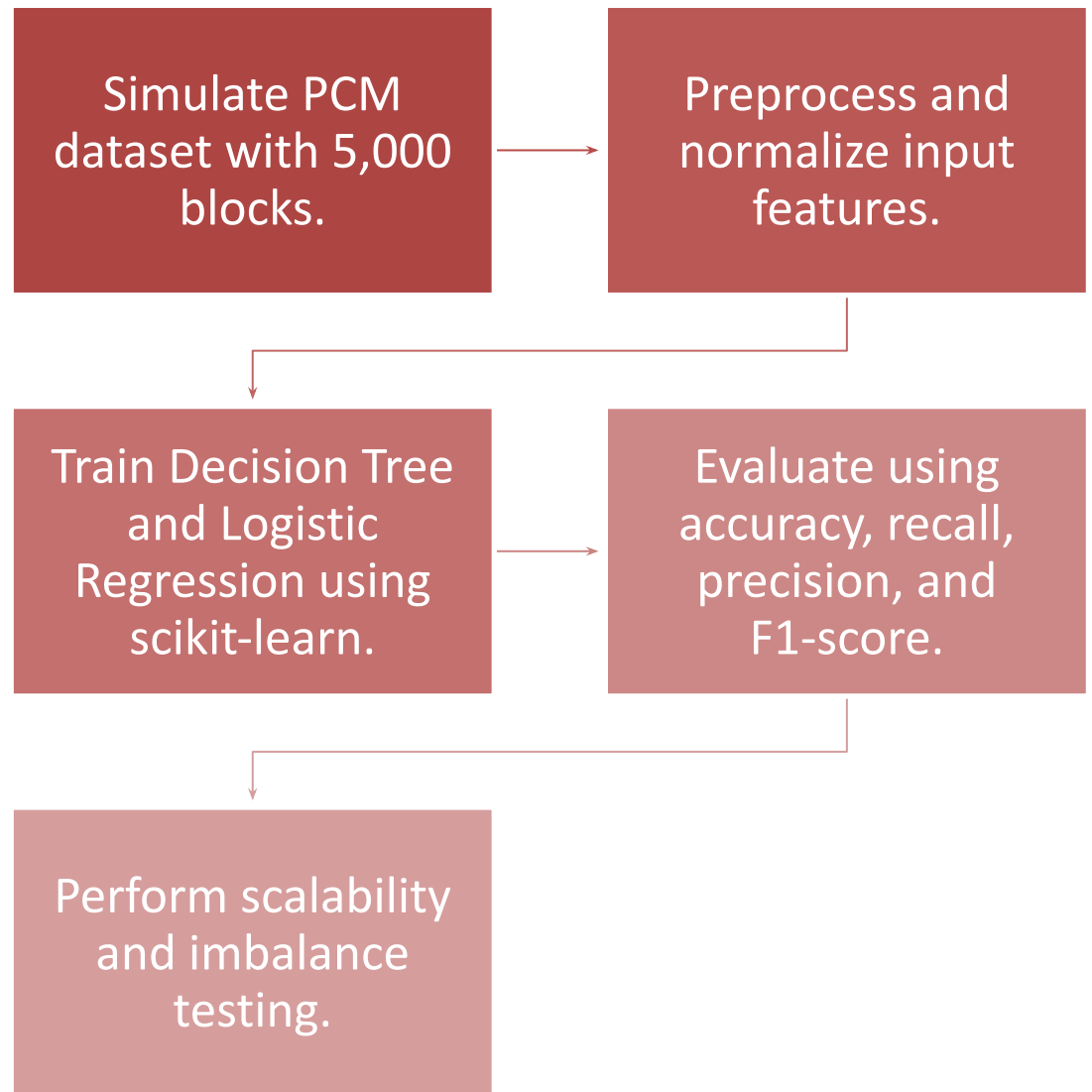
Prior studies use deep learning models—accurate but resource-heavy.

Limited research on lightweight interpretable models.

Our approach: Decision Tree and Logistic Regression for simplicity and transparency.

Balance between accuracy, interpretability, and computational cost.

Methodology Overview



Dataset Collection

5,000 simulated PCM memory blocks.

Features: Write Count (1K–80K), Temperature (25–100°C), Error Frequency.

Balanced dataset: 65% Healthy, 25% Degraded, 10% Near-Failure.

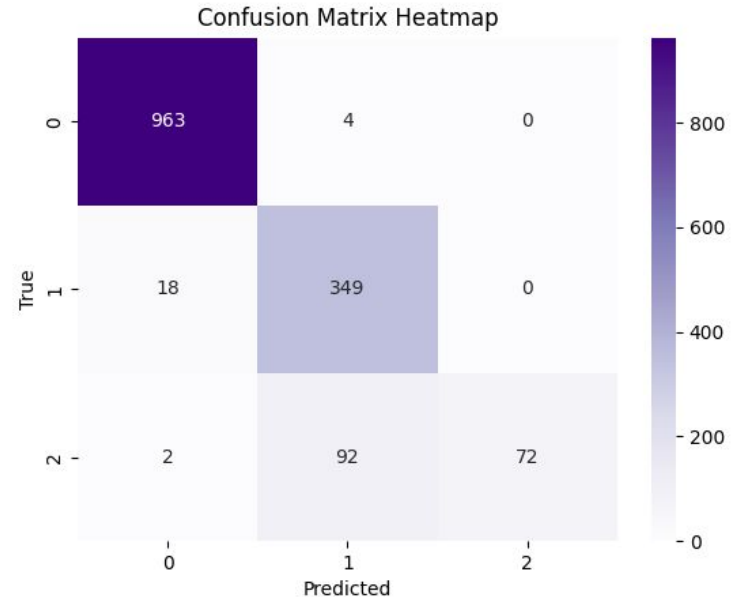
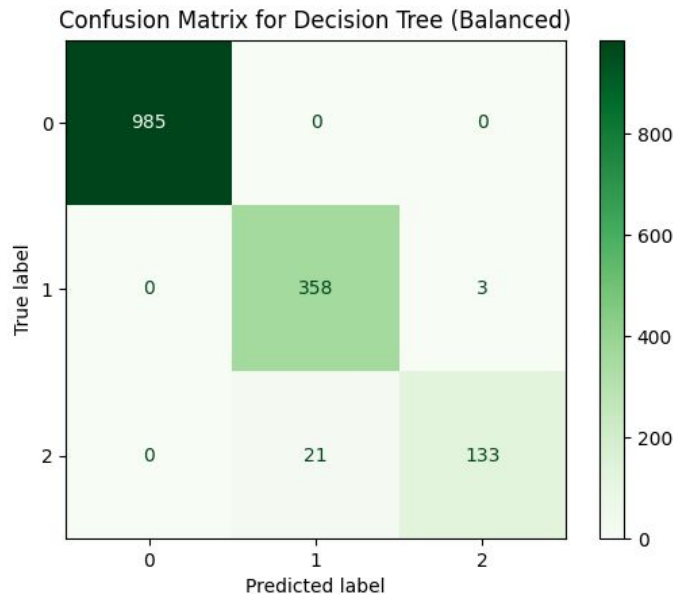
Imbalanced dataset: more Degraded cells to simulate realistic wear

Model Setup

- Implemented in Python (scikit-learn, pandas, matplotlib).
- 80/20 train-test split, 5-fold cross-validation.
- Decision Tree tuned via GridSearchCV (max_depth, min_samples_split).
- Logistic Regression tuned (C, penalty type) using saga solver.
- Evaluated with confusion matrices and performance metrics.

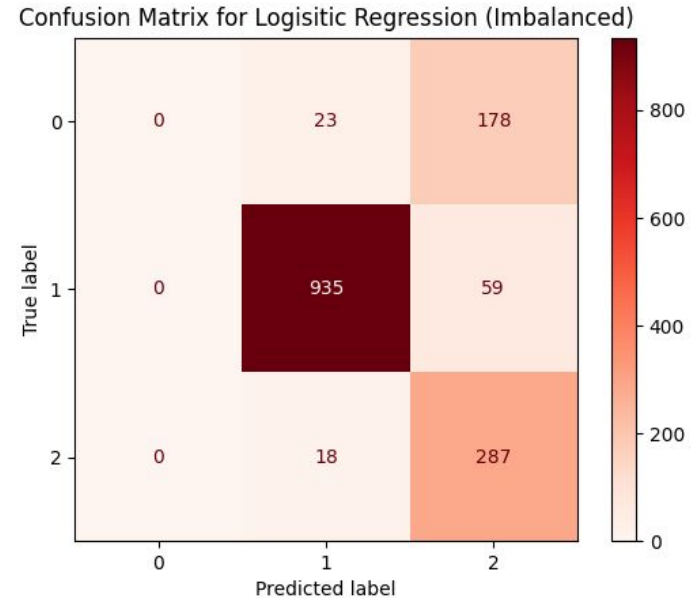
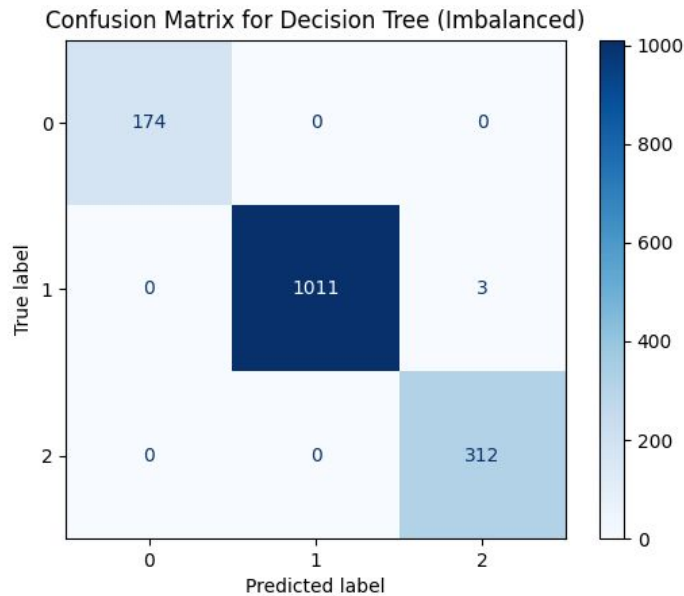
Results: Balanced Dataset

- Decision Tree achieved 98.4% accuracy.
- Logistic Regression achieved 92% accuracy.
- Near-Failure detection weaker for Logistic Regression (Recall = 0.43).
- Decision Tree shows robust class separation and interpretability.



Results: Imbalanced Dataset

- Decision Tree: 100% accuracy with strong minority class detection.
- Logistic Regression: 81% accuracy; failed to detect Healthy class.
- Indicates Decision Trees handle class imbalance more effectively.



Feature Importance

- Top predictors: Write Count, Temperature, and Error Frequency.
- High write count and elevated temperature → Near-Failure cells.
- Feature importance varies between balanced and imbalanced datasets.

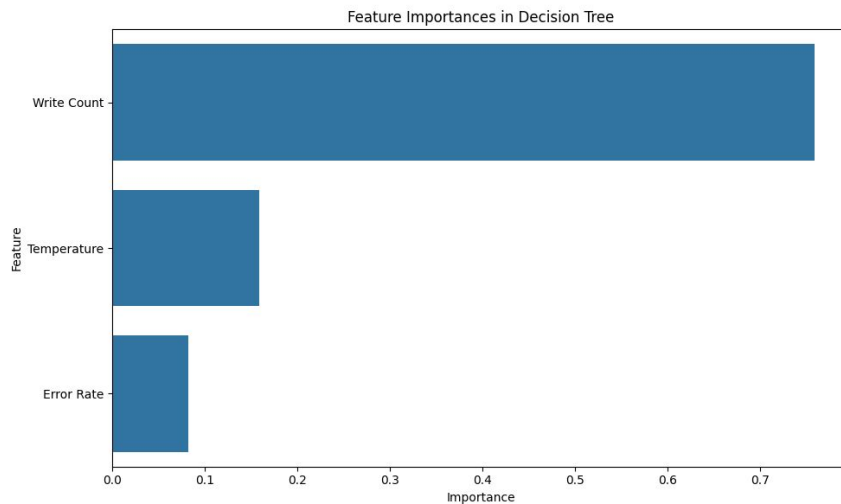


Figure 1: Balanced Feature Importance

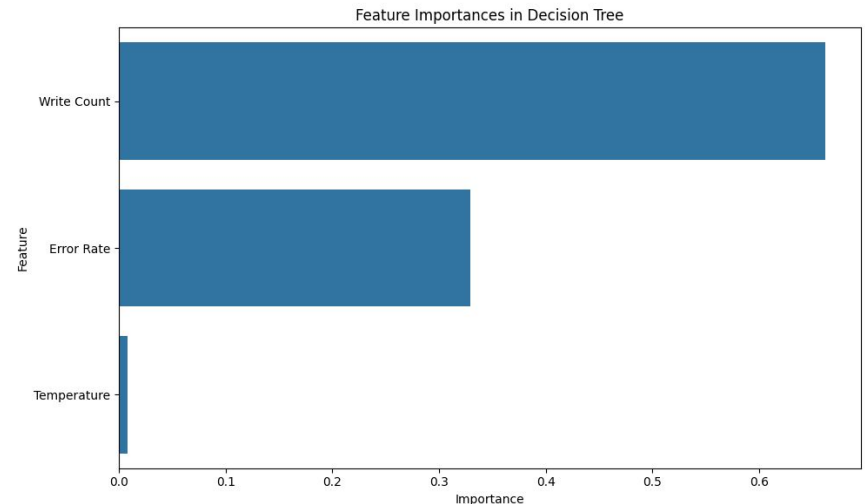


Figure 2: Imbalanced Feature Importance

Scalability Analysis

- Evaluated on 7.5K–50K memory blocks.
- Accuracy improved and stabilized with larger datasets.
- Balanced datasets consistently yielded better results.
- Confirms robustness of Decision Tree for large-scale PCM monitoring.

Conclusion and Future Work

- Lightweight models provide efficient and interpretable PCM health classification.
- Decision Tree > Logistic Regression for handling imbalance and nonlinearities.
- Supports real-time, low-overhead wear monitoring in embedded systems.
- Future Work:
 - Validate on real PCM hardware.
 - Integrate with wear-leveling firmware.
 - Explore ensemble and explainable AI approaches.